

FlockDefender: An IoT Based Poultry House Monitoring and Heat Stress Prediction System for a Poultry Farm in San Pablo, Santa Ana, Pampanga

Rationale

Many poultry farmers in the Philippines struggle with keeping their chickens healthy because of sudden changes in weather. Chickens are very sensitive to heat and bad smells like ammonia from their waste. When it gets too hot or the air gets dirty, chickens can get sick or even die, which causes farmers to lose a lot of money. In the poultry farm located in San Pablo, Santa Ana, Pampanga, the caretakers usually check the coop manually. They do not use thermometers or any special tools to measure the environment. They just feel the air or wait until the chickens look tired or sick.

Traditional ways of checking the farm rely on guessing, but this is often too late to save the chickens. This study proposes FlockDefender, a smart system that uses sensors to watch over the poultry house day and night. The system monitors the environment and sends a warning to the owner immediately if there is a risk. It helps the farm owner know exactly when the temperature or air quality is getting dangerous so they can act fast before problems happen.

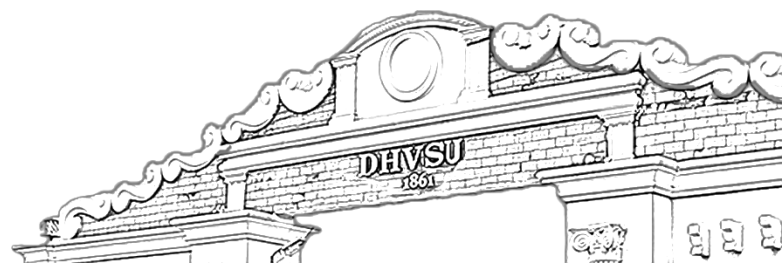
Project Purpose

The purpose of this project is to create a unified digital system that improves how the environment is managed at the poultry farm. By using sensors and a mobile app, the owner can see the real status of the coop anytime. This helps keep the chickens safe and gives the owner the power to control the cooling fans automatically or manually through their phone.

Research Objectives

- To build a hardware prototype with sensors that can measure temperature, humidity, and ammonia levels 24 hours a day.
- To create a mobile app that shows the real time status of the poultry house using simple colors like green for safe, yellow for warning, and red for danger.
- To develop a system that sends a stress alert to the user's phone when it detects that the air conditions are becoming dangerous for the chickens.
- To integrate a fan control module that can turn on the cooling fans automatically when it is hot or let the user turn them on and off manually through the app.

Scope and Limitation of the Study



Scope

The system will focus on tracking air temperature, humidity, and ammonia gas inside the poultry house in Santa Ana. It will include a mobile app and a website where the owner can receive alerts and view graphs of the farm conditions. The hardware will use a relay module to let the user control the cooling fans both automatically and manually.

Limitations

- The system does not check for viruses or give medical advice for sick chickens.
- The alerts and controls are only based on environmental factors like heat and gas levels.
- The app needs a working internet connection or local Wi-Fi at the farm to send data and receive manual commands.

Benefits/Anticipated Outcomes

The Farm Owner: It helps them avoid losing money by giving them a way to monitor and control their farm even when they are away.

The Caretaker: It makes their job easier because the system monitors the coop automatically, so they do not have to rely on guessing the temperature or air quality.

Methodology

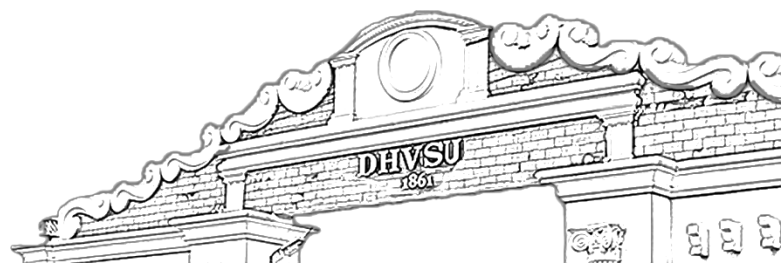
This study will use the Agile Software Development Framework to make sure the system is built and improved properly.

Requirement Gathering: We will talk to the farm owner and staff to understand their common problems with the weather and fan settings.

System Development: We will design and code the mobile app and put together the sensor hardware including the manual control features.

Testing and Deployment: The system will undergo testing at the farm to make sure the sensors are accurate and the fans respond to the app commands.

Evaluation and Improvement: We will collect feedback from the user to make the app easier to use and more reliable for the farm.



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